■■ **DUMMY / HACKATHON SIMULATION RECORD: This is a hackathon/training simulation and does not constitute a regulatory audit, validation decision, or production release authorization.**

GxP IT AUDIT & LIFECYCLE INTELLIGENCE REPORT

System: Novo Life MES PAS-X (SYS-MES-001) | Checklist: Top 25 Difficult-Auditor Questions

1. Executive Summary

An independent audit evaluation was conducted on **Novo Life MES PAS-X (SYS-MES-001)** using the **Top 25 Difficult-Auditor GxP IT Checklist (2026)**, benchmarked against **NN Master IT System Lifecycle SOP (HACK-IT-SOP-001)** and the **GxP LIMS Lifecycle Documentation Package (LIMS-LCP-001)**. The calculated **Overall Audit Readiness Score is 64.6%**. Due to critical incomplete verification activities and unrated residual risks, the formal system release recommendation remains: **HOLD / DEFER - DO NOT RELEASE TO PRODUCTION**.

Readiness Score	Total Questions	Passed	Partial	Failed	Critical Findings	Release Decision
64.6%	25	15	4	6	4	HOLD / DEFER

2. System Assessed & 3. Assessment Metadata

System ID / Name	SYS-MES-001 | Novo Life MES PAS-X (Werum PAS-X Category 4)
Current Status	PRE-OPERATIONAL / NOT ACTIVATED [NL-MES-SLA-001 p.1]
Assessment Date	2026-09-03 19:31:31 UTC
Auditing Engine	GxP IT Audit & Lifecycle Intelligence Assistant (Deterministic Engine)
Predicate Rules	21 CFR Part 11, 21 CFR 211.68, EU GMP Annex 11 / 15, GAMP 5, ICH Q9(R1)

4. Documents Reviewed Across Knowledge Hierarchy

Scope	Document ID	Title	Version	Role / Status
Primary Evidence	NL-MES-URS-001	User Requirements Specification	v1.0	Draft / Missing QA Approval
Primary Evidence	NL-MES-MLGP-001	Master Lifecycle Generation Plan	v1.0	Approved (Simulated)
Primary Evidence	NL-MES-ITRA-001	IT Risk Assessment Baseline	v1.0	Approved (Simulated)
Primary Evidence	NL-MES-ITRRA-001	Requirement Risk Assessment	v1.0	Residual Risk NOT RATED
Primary Evidence	NL-MES-IREP-001	IT Implementation Report	v1.0	Gate G5 Blocked / OV Open
Primary Evidence	NL-MES-ITPSE-001	Periodic System Evaluation	v1.0	HOLD / DEFER Conclusion
Primary Evidence	NL-MES-SLA-001	Service Level Agreement	v0.9	Pre-Operational / Unactivated
Governance SOP	HACK-IT-SOP-001	Manage IT System Lifecycle SOP	v0.1	Master Governance Standard
Benchmark Ref	LIMS-LCP-001	GxP LIMS Lifecycle Package	v0.1	Laboratory Benchmark Ref
Audit Checklist	CKL-TOP25-2026	Top 25 Difficult-Auditor Questions	2026.1	Master Audit Instrument

5 & 6. Audit Checklist: Question-by-Question Results

#	Q_ID	Audit Domain / Topic	Result	Evidence Citation	Severity

1	DA-01-001	**Intended Use & System Boundary** Walk through the proposed intended use, patient/product decisions supported...	PASS	[NL-MES-MLGP-001 | p.1 | System Scope & Classification]	LOW
2	DA-01-003	**Data Lifecycle & ALCOA+ Protection** Describe the data lifecycle and show where the concept explicitly protects ...	PASS	[NL-MES-URS-001 | p.2 | Section 2.1 Data Integrity Controls]	LOW
3	DA-02-001	**Completeness of GxP User Requirements** Demonstrate that the URS captures all GxP, operational, data integrity, and...	PARTIAL	[NL-MES-URS-001 | p.1 | Executive Summary]	HIGH
4	DA-03-001	**Initial System Risk Assessment & Hazard Identification** Show how system risk assessment differentiated patient-safety, product-qual...	PASS	[NL-MES-ITRA-001 | p.2 | Risk Register RSK-MES-001 to 026]	LOW
5	DA-03-005	**Authorized Residual Risk Acceptance** Demonstrate that all high-risk items have verified mitigations and that res...	FAIL	[NL-MES-ITRRA-001 | p.3 | Section 3 Residual Risk Evaluation]	CRITICAL
6	DA-04-001	**Supplier Audit & Quality Agreement** Confirm that the software supplier (Werum IT Solutions) underwent formal au...	PASS	[NL-MES-SUPA-001 | p.1 | Werum Supplier Audit & Qualification]	LOW
7	DA-07-001	**Technical Environment & Installation Verification** Verify that all production hardware, operating systems, database schemas, a...	PASS	[NL-MES-IREP-001 | p.2 | Section 3.1 Technical Verification Summary]	LOW
8	DA-08-001	**Functional & Security Control Verification** Show that all automated functions, calculations, security permissions, and ...	PASS	[NL-MES-IREP-001 | p.2 | Section 3.1 Technical Verification]	LOW
9	DA-09-001	**Intended-Use Qualification & Shopfloor Verification (OV / PfV / UAT)** Demonstrate that the system was tested under realistic operating conditions...	FAIL	[NL-MES-IREP-001 | p.2 | Section 3.2 Intended-Use Verification Gap]	CRITICAL
10	DA-10-001	**Operational Handover, Service Level Agreements & Support Readiness** Show that operational support models, SLA tiers, incident response escalati...	FAIL	[NL-MES-SLA-001 | p.1 | Section 1 System Operational Status]	HIGH
11	DA-10-005	**End-User Training & Qualification Records** Demonstrate that all personnel with access to the system are trained on rel...	FAIL	[NL-MES-SLA-001 | p.2 | Appendix A Training Matrix]	HIGH
12	DA-10-007	**Validation Summary Report (VSR) & Formal Release Gate G5** Confirm that a Validation Summary Report synthesizing all qualification res...	FAIL	[NL-MES-IREP-001 | p.4 | Section 4.3 Validation Summary Report]	CRITICAL
13	DA-11-001	**Periodic System Evaluation & Validated State Conclusion** Reconstruct the most recent periodic evaluation from source populations and...	PARTIAL	[NL-MES-ITPSE-001 | p.1 | Periodic System Evaluation]	HIGH
14	DA-11-005	**Backup, Disaster Recovery & Business Continuity Testing** Walk through the most recent backup and restore test and verify that disast...	PASS	[NL-MES-IREP-001 | p.2 | Section 3.1 Technical Verification]	LOW
15	DA-12-001	**Production Change Population & Reconciliation** Build the complete population of production changes and reconcile it to app...	PARTIAL	[NL-MES-MLGP-001 | p.2 | Change Management Section]	MEDIUM

16	DA-12-005	**Emergency Change & Post-Implementation Review** Examine the population of emergency and hotfix changes to ensure that retro...	PASS	[NL-MES-MLGP-001 | p.3 | Section 4 Emergency Change Protocol]	LOW
17	DA-13-001	**Event Population & GxP Impact Assessment** Reconcile system monitoring alerts, service desk tickets, and audit trail e...	PARTIAL	[NL-MES-SLA-001 | p.2 | Incident Triage Procedures]	MEDIUM
18	DA-13-005	**CAPA Linkage & Root Cause Analysis** Trace recurrent incidents or critical defects to root-cause investigation a...	PASS	[NL-MES-MLGP-001 | p.3 | CAPA Management Procedures]	LOW
19	DA-02-005	**Audit Trail & Electronic Signature Compliance (Part 11 / Annex 11)** Show that the audit trail is secure, computer-generated, time-stamped, and ...	PASS	[NL-MES-URS-001 | p.3 | URS-028 Audit Trail Display]	LOW
20	DA-05-001	**Functional Specification & Architecture Modularity** Demonstrate that functional specifications trace directly to user requireme...	PASS	[NL-MES-URS-001 | p.2 | Section 2]	LOW
21	DA-06-001	**Configuration Management & Repository Control** Verify that all configuration files, master recipe parameters, and source c...	PASS	[NL-MES-MLGP-001 | p.2 | Lifecycle Governance]	LOW
22	DA-14-001	**Data Retention, Archival & Retrieval Readability** Show that electronic batch records and audit trails can be retained and ret...	PASS	[NL-MES-MLGP-001 | p.3 | Section 5 Archival & Decommissioning]	LOW
23	DA-02-010	**Role-Based Access Control & Segregation of Duties** Verify that user authorization enforces least privilege, prevents self-appr...	PASS	[NL-MES-URS-001 | p.2 | URS-009 Role-Based Access]	LOW
24	DA-08-010	**Automated Interface Verification (ERP / SCADA / Serialization)** Demonstrate that data exchanges between MES, SAP ERP, and shopfloor packagi...	PASS	[NL-MES-IREP-001 | p.2 | Section 3.1 Interface Testing]	LOW
25	DA-10-025	**Release Gate Checklist Reconciliation & Regulatory Readiness** Reconcile all lifecycle gate deliverables (G1 through G6) to verify no unap...	FAIL	[NL-MES-IREP-001 | p.3 | Gate G1-G6 Summary]	CRITICAL

7 & 8. Critical Audit Findings & Evidence

FINDING [DA-02-001]: Audit Finding [DA-02-001]: Completeness of GxP User Requirements **Severity: HIGH**

Gap: Evidence indicates Quality Unit approval signature remains pending on the baseline URS document.
Recommendation: Route URS-001 through formal electronic Quality Unit sign-off.
Evidence: [NL-MES-URS-001 | p.1 | Executive Summary], [NL-MES-ITRRA-001 | p.1 | Section 1], [Top_25_Checklists_GxP_IT_Audit_Questions_2026.xlsx | 02 URS | Row 5]

FINDING [DA-03-005]: Audit Finding [DA-03-005]: Authorized Residual Risk Acceptance **Severity: CRITICAL**

Gap: Residual risk evaluation is unrated and unapproved for 49 critical working requirements.
Recommendation: Conduct an authorized residual risk review session with the Quality Unit to sign off on working high risks.
Evidence: [NL-MES-ITRRA-001 | p.3 | Section 3 Residual Risk Evaluation], [HACK-IT-SOP-001 | p.12 | Section 6.2 Risk Scales and Acceptance], [Top_25_Checklists_GxP_IT_Audit_Questions_2026.xlsx | 03 Risk GAMP | Row 9]

FINDING [DA-09-001]: Audit Finding [DA-09-001]: Intended-Use Qualification & Shopfloor Verification (OV / PfV / UAT) **Severity: CRITICAL**

Gap: Operational verification on commercial packaging lines was deferred and remains unperformed, directly invalidating release readiness.
Recommendation: Execute intended-use qualification test scripts on the packaging line before seeking Gate G5 release authorization.
Evidence: [NL-MES-IREP-001 | p.2 | Section 3.2 Intended-Use Verification Gap], [NL-MES-IREP-001 | p.3 | Section 4.1 Gate G5 Status], [HACK-IT-SOP-001 | p.17 | Section 7.2 Verification Execution], [Top_25_Checklists_GxP_IT_Audit_Questions_2026.xlsx | 09 PQ UAT | Row 5]

FINDING [DA-10-001]: Audit Finding [DA-10-001]: Operational Handover, Service Level Agreements & Support Readiness **Severity: HIGH**

Gap: Support handover is not executed; Release Gate G6 is marked NOT MET.
Recommendation: Activate IT service management SLA tiers and finalize operational support agreements.
Evidence: [NL-MES-SLA-001 | p.1 | Section 1 System Operational Status], [NL-MES-IREP-001 | p.4 | Section 4.2 Gate G6 Status], [Top_25_Checklists_GxP_IT_Audit_Questions_2026.xlsx | 10 GoLive Handover | Row 5]

9. Cross-Document Comparison & Lifecycle Deviations

Direct comparison of MES PAS-X evidence against Master IT SOP (HACK-IT-SOP-001):

Topic	SOP Expectation	MES PAS-X Observed	Status	Recommended Action
Intended-Use Verification & Performance Qualification	Prior to commercial release authorization (Gate G5), the system must undergo documented intended-use qualifica...	NL-MES-IREP-001 Section 3.2 records operational verification (OV / PfV / UAT) as NOT PERFORMED....	POTENTIAL LIFECYCLE DEVIATION	Execute intended-use qualification test scripts on commercial packaging lines prior to reconven...
Residual Risk Quality Unit Acceptance	Risk-based lifecycle assurance requires all critical requirements to have verified mitigations and explicit, s...	NL-MES-ITRRA-001 indicates residual risk is NOT RATED across 49 working high requirements....	POTENTIAL LIFECYCLE DEVIATION	Conduct formal residual risk acceptance evaluation with the Quality Unit to review the 49 worki...
Operational SLA Handover & Operator Training	Operational handover (Gate G6) mandates signed Service Level Agreements, activated incident management tiers, ...	NL-MES-SLA-001 is in PRE-OPERATIONAL / NOT ACTIVATED status; 0 of 250 packaging operators are trained....	POTENTIAL LIFECYCLE DEVIATION	Qualify 250 operators on MES packaging SOPs and transition SLA from Pre-Operational to Active....
Installation Qualification (IQ) & Technical Environment	Technical components, database schemas, and interface middleware must be verified against design specification...	NL-MES-IREP-001 Section 3.1 records technical integration, infrastructure qualification, and database installa...	ALIGNED	Maintain automated checksum monitoring on infrastructure configuration....

11 & 12. Corrective Actions Required Prior to Production Release

1. **Execute Intended-Use Verification (OV / PfV / UAT):** Complete qualified test scripts on packaging lines to unblock Gate G5 [NL-MES-IREP-001 p.2].

2. **Conduct Authorized Residual Risk Review:** Review and accept the 49 working high risks with the Quality Unit [NL-MES-ITRRA-001 p.3].

3. **Finalize and Authorize Validation Summary Report (VSR):** Route the VSR for formal QA sign-off prior to Gate G5 authorization.

4. **Complete Shopfloor Operator Training & SLA Activation:** Train 250 packaging line operators and activate 24/7 SLA operational support [NL-MES-SLA-001 p.1].

13 & 14. Limitations & Regulatory Disclaimer

LIMITATIONS: This is a hackathon/training simulation and does not constitute a regulatory audit, validation decision, or production release authorization. All findings, scores, and release recommendations are deterministic outputs generated from synthetic documentation for hackathon demonstration purposes.